LECTURE 08 · REVIEW NOTES

Where Did the Cash Go?

Free cash flow to equity, free cash flow to the firm, and the bridge from operations to common equity.

Luis Ceballos, PhD · FINA 402 · Fall 2026

1. Profit, cash flow and distributions

Lecture 07 valued expected dividends. Lecture 08 estimates the cash available to capital providers after operating investment. Dividend policy describes distributions; cash generation describes the ability to fund them. A firm can create value while paying little or no current dividend.

Microsoft FY2025 · $ billions	Amount
Net income	101.832
Net cash from operations	136.162
Cash additions to property and equipment	64.551
Operating cash flow less cash CapEx	71.611

Fiscal year ended June 30, 2025. The last row is our subtraction of two reported figures. It is a useful introductory cash measure, not a complete FCFE estimate. Other investments, financing flows, leases and noncash adjustments require consistent treatment in a full valuation.

Source: Microsoft FY2025 cash flow statement. Historical reported data, not a forecast.

Free cash flow to equity

$$FCFE = NI + D\&A - CapEx - \Delta NWC + \text{Net debt issued}$$

NI is net income after interest and taxes. D&A reverses a noncash expense already deducted from earnings. Cash capital expenditure uses funds for long-term assets. ΔNWC is the increase in noncash operating working capital: an increase uses cash, while a decrease releases it. Net debt issued equals new borrowing minus principal repayments.

Debt issuance can increase this period’s FCFE, but creates future obligations. The classroom debt control isolates the current cash effect. It does not establish that borrowing creates wealth or that the same discount rate remains appropriate under different leverage.

LECTURE 08 · REVIEW NOTES · 2

2. From FCFE to equity value

Illustrative Manufacturing has the following current-year amounts. All dollar inputs are in millions. These are practice data, not a student's graded assignment.

Item	Amount	Treatment
Net income	100	Starting point
Depreciation and amortization	20	Add back
Cash capital expenditure	70	Subtract
Increase in noncash operating WC	15	Subtract
Net debt issued	5	Add

$$FCFE_0 = 100 + 20 - 70 - 15 + 5 = 40$$

An equivalent calculation uses net CapEx = 70 − 20 = 50. Equity reinvestment is 50 + 15 − 5 = 60, so FCFE = 100 − 60 = 40. Do not add depreciation twice.

Stable-growth valuation

$$FCFE_1 = FCFE_0(1 + g)$$
$$Equity_0 = FCFE_1 / (k_e - g)$$

This formula assumes sustainable stable growth, a consistent investment and financing policy, and k_e greater than g. Both rates must be expressed as decimals. A temporary or negative cash flow is not automatically an appropriate perpetual base.

With g = 3%, k_e = 11%, and 20 million shares:

Step	Result
Next-year FCFE: 40 × 1.03	$41.2m
Equity value: 41.2 / (0.11 − 0.03)	$515m
Price: 515 million / 20 million shares	$25.75 per share

Do not subtract debt again. FCFE is already the cash flow available to equity after the financing adjustments. In this simplified example there are no separately valued nonoperating assets.

Sensitivity

Holding FCFE and growth fixed, increasing k_e to 12% changes equity value to 41.2/0.09 = $457.78m, or $22.89 per share. Explain the direction before calculating. A more complete change in financing policy can also change cash flows and risk.

LECTURE 08 · REVIEW NOTES · 3

3. Value operations, then common equity

FCFF = EBIT(1 − T) + D&A − CapEx − ΔNWC

FCFF is operating cash flow available to debt and equity providers. Start with earnings before interest, apply operating taxes, and deduct reinvestment. Net borrowing is excluded. Use WACC to discount FCFF, while FCFE uses the cost of equity.

$WACC = w_e k_e + w_d k_d(1 - T)$

Financing weights use market values and sum to one. The simple formula assumes the modeled debt tax benefit is available. The activities supply WACC so you can focus on cash flows and the equity bridge.

Worked example · all amounts in millions

EBIT = 120; tax = 25%; D&A = 20; CapEx = 40; ΔNWC = 10. Stable growth is 3% and WACC is 9%. Debt is 200, excess cash is 50, and there are 40 million common shares. No other claims or nonoperating assets are assumed.

Step	Calculation	Result
After-tax operating income	120 × (1 − 0.25)	90
Current FCFF	90 + 20 − 40 − 10	60
Next-year FCFF	60 × 1.03	61.8
Operating value	61.8 / (0.09 − 0.03)	1,030
Common equity	1,030 − 200 + 50	880
Price per share	880 / 40	$22.00

Common equity = Operating value − Debt + Excess cash

In practice, adjust for other claims such as preferred equity and noncontrolling interests, and for separately valued nonoperating assets. Add only cash excluded from operating value. Counting required operating cash twice overstates equity.

The slide controls hold operating value fixed and change debt or excess cash to illustrate the bridge. Increasing debt from 200 to 300, holding everything else fixed, lowers equity to 780 and price to $19.50. This is a comparison of balance-sheet claims, not a simulation of issuing debt and retaining its cash proceeds.

LECTURE 08 · REVIEW NOTES · 4

4. Your activities and review checklist

Activity	Outputs	Points
1 · FCFE valuation	Current FCFE and total equity value ($m)	3
2 · FCFF valuation	Operating value ($m) and price per share ($)	2

Each student receives a distinct, repeatable teaching case. Both outputs must be correct to complete an activity. Tolerance is ±1%; keep intermediate precision. Complete Activity 1 before Activity 2. Correct saved results appear on the instructor's section dashboard. Demonstrations do not create submissions. Review mode disables graded submissions.

Before submitting

Check	Reason
Current or next-year cash flow?	Grow current cash flow once for the perpetual numerator.
What does ΔNWC mean?	An increase uses cash. Subtracting a negative change adds cash.
Whose cash flow?	FCFE uses cost of equity. FCFF uses WACC.
What value did I calculate?	FCFE gives equity directly. FCFF gives operating value.
Which units?	Millions of dollars divided by millions of shares gives dollars per share.
Is perpetual growth sustainable?	Require the discount rate to exceed g; align growth with reinvestment and financing.

Self-check

1. NI = 80, D&A = 10, CapEx = 30, ΔNWC = −5, and net debt = −15. Calculate FCFE.
2. Using that FCFE, g = 2% and k_e = 12%, calculate total equity value.
3. Operating value is 900, debt is 200, excess cash is 50, and shares are 25. Calculate the stock price.
4. Why is subtracting debt after an FCFE valuation generally a mistake?

Solutions. (1) 80 + 10 − 30 − (−5) + (−15) = **50**. (2) 50 × 1.02 / (0.12 − 0.02) = **510**. (3) (900 − 200 + 50) / 25 = **$30**. (4) FCFE already measures cash available to equity after financing; subtracting debt again double counts the adjustment.

Discussion prompts

Why can profitable firms have negative FCFE? Why can borrowing increase current FCFE without creating value? If two analysts use FCFE and FCFF consistently, what assumptions must agree for their equity valuations to reconcile?